

Problem-Solution Fit

Date	26 June 2026
Team ID	SWTID-2026-8797
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

1. CUSTOMER SEGMENT(S) [CS] • Banks and financial institutions • Credit analysts and loan officers • Customers applying for credit cards	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] • Limited financial information • Incomplete application details • Varying income sources • Limited credit history • Dependence on manual processing	5. AVAILABLE SOLUTIONS PROS & CONS [AS] • Manual Verification: Accurate but slow and costly • Traditional Rule-Based Systems: Fast but lacks adaptability • Machine Learning Models: Fast, scalable, accurate and learns from historical data
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] • Manual verification is slow and time-consuming • Human errors during evaluation • Large number of applications received daily • Delays and inconsistent approval decisions	9. PROBLEM ROOT / CAUSE [RC] • Increasing application volume • Manual verification process • Inconsistent human decision-making • Incomplete applicant data • Inefficient risk assessment	7. BEHAVIOR + ITS INTENSITY [BE] • Applicants expect quick approval decisions • Banks process thousands of applications daily • Need for high-speed, consistent and reliable evaluation
3. TRIGGERS TO ACT [TR] • Increase in credit card applications • Need for faster approval • Customer demand for instant decisions • Requirement to reduce operational costs	10. YOUR SOLUTION [SL] Develop a Credit Card Approval Prediction System using Machine Learning (Logistic Regression, Decision Tree, Random Forest and XGBoost) integrated with a Flask web application. The system predicts approval or rejection instantly based on applicant information, improving accuracy, reducing processing time and assisting banks in making reliable decisions.	8. CHANNELS OF BEHAVIOR [CH] ONLINE • Bank websites • Mobile banking applications • Online credit card portals OFFLINE • Bank branches • Customer service centers • Financial consultants
4. EMOTIONS BEFORE / AFTER [EM] Before: Frustration, uncertainty, long waiting time and stress After: Faster approval, confidence, transparency, improved customer satisfaction and trust		